

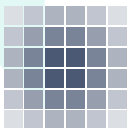
CSE220 • SIGNALS AND LINEAR SYSTEMS SESSIONAL

Fourier-Domain Image Encryptor

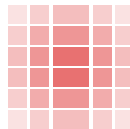
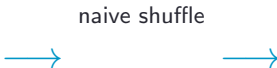
Md. Rijwanullah Sayem • Al-Amin Md. Saifullah

Supervisor: Junaed Younus Khan • CSE, BUET

Why Is Image Encryption Hard?



Plaintext: correlated pixels

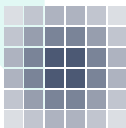


Outline still leaks

► The core challenge

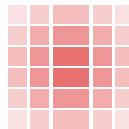
- Images $\neq$ text: neighbouring pixels are strongly **spatially correlated**

Why Is Image Encryption Hard?



Plaintext: correlated pixels

naive shuffle

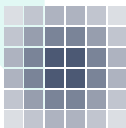


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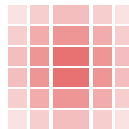
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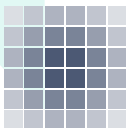


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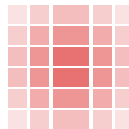
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Pixel-domain scrambling alone is not enough



01 – STANDARD DRPE

The Solution:
Move to the Frequency Domain

Mixes every ciphertext pixel
with the whole image, removing
direct spatial correspondence

➤ Why it works

Standard D



Mixes every ciphertext pixel
with the whole image, removing
direct spatial correspondence

Two phase masks scramble magnitude and phase in separate domains

► Why it works

Standard DRPE — The Idea



2D FFT (built from scratch)

Mixes every ciphertext pixel
with the whole image, removing
direct spatial correspondence



Two Random Phase Masks

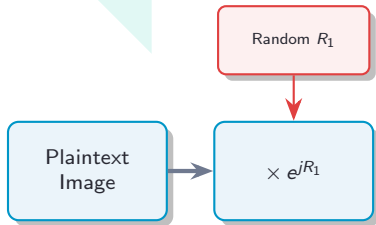
Two phase masks scramble magnitude and phase in separate domains

➤ Why it works

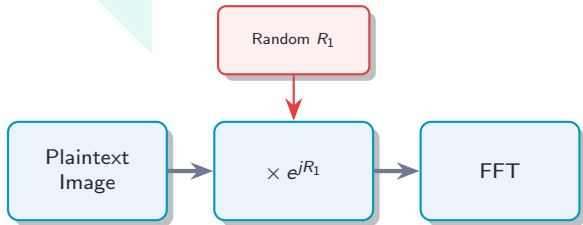
- Frequency-domain energy from every pixel is spread across the *whole* ciphertext
- Result looks like pure noise — no visible outline of the original image
- Historically realized optically with lenses & ground-glass diffusers

Plaintext
Image

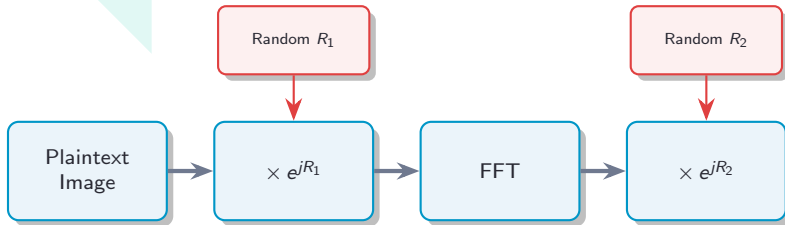
Standard DRPE — Flow



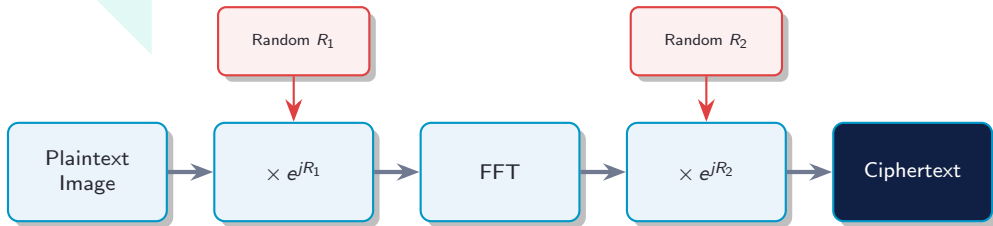
Standard DRPE — Flow



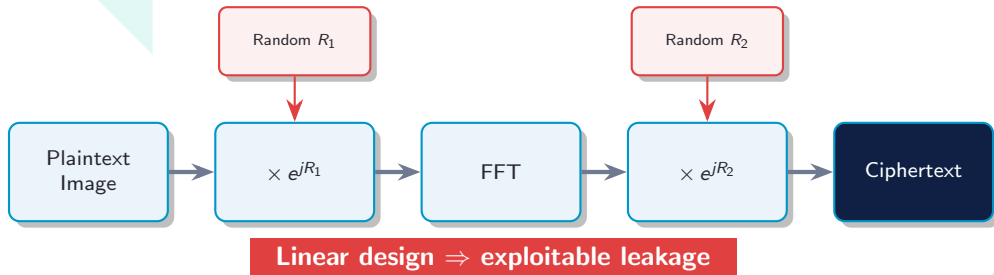
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Standard DRPE — Flow



Standard DRPE — Flow



Naive scrambling
shapes & edges still visible

➤ The flaw in what you just saw

Vulnerability: Why It Fails



Naive scrambling

shapes & edges still visible



$\times R_1 \rightarrow \text{FFT} \rightarrow \times R_2$

purely *linear* $\Rightarrow$ small
key error = small error

image still leaks

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► The flaw in what you just saw

- Neighboring pixels are correlated
- Attackers reverse-engineer near-correct keys
- Chosen-Plaintext Attacks expose static keys

Vulnerability: Why It Fails



Naive scrambling

shapes & edges still visible



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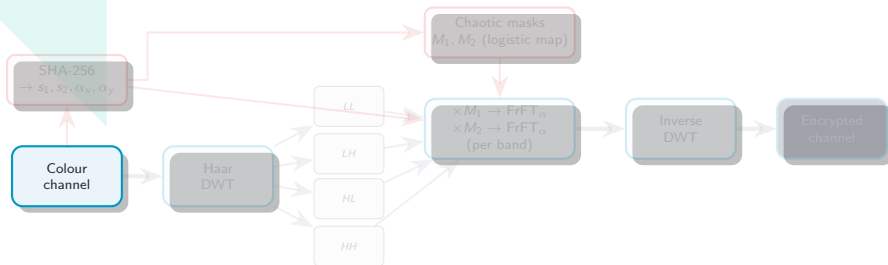
**Goal: total structural destruction + perfect
legitimate recovery**



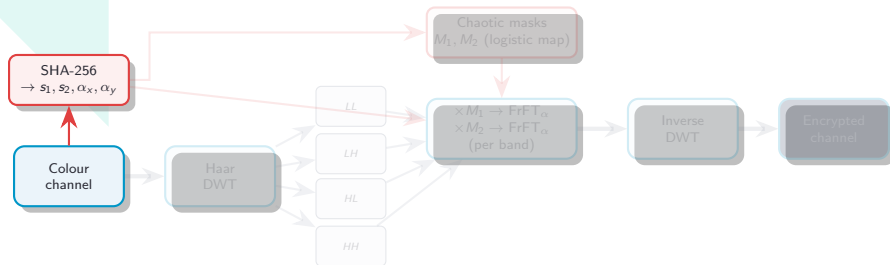
02 – ADVANCED PIPELINE

Fixing the Flaw: Chaotic DRPE

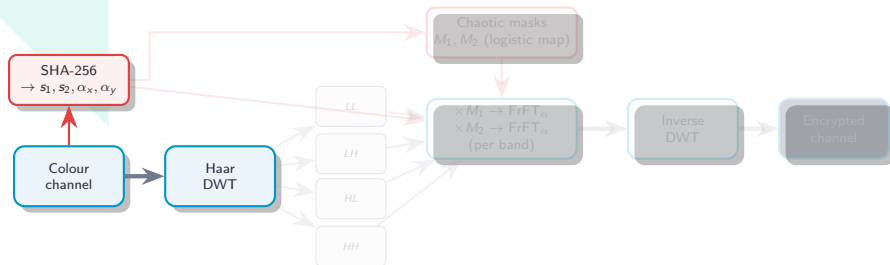
Advanced Chaotic DRPE — Flow



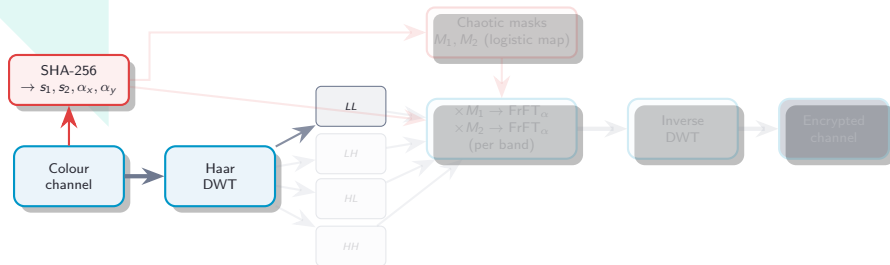
Advanced Chaotic DRPE — Flow



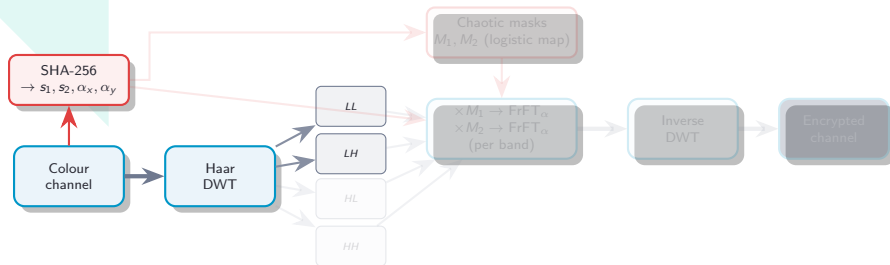
Advanced Chaotic DRPE — Flow



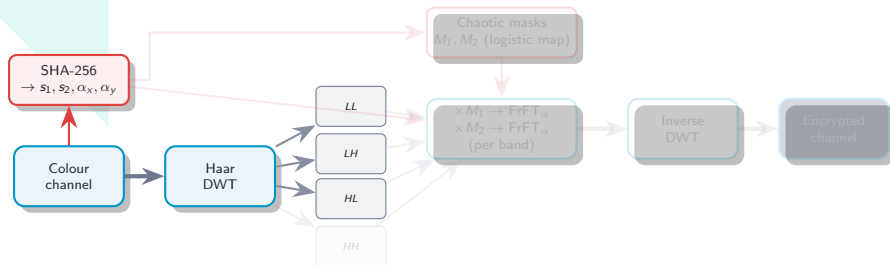
Advanced Chaotic DRPE — Flow



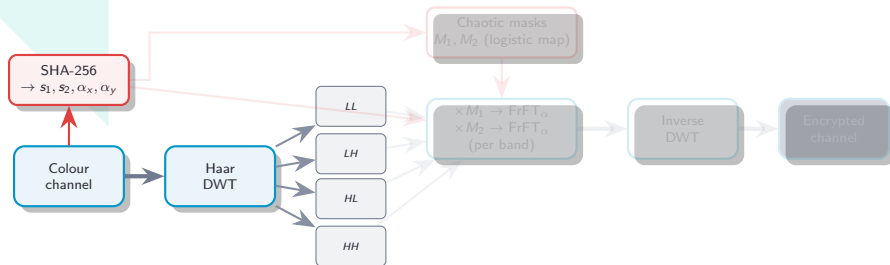
Advanced Chaotic DRPE — Flow



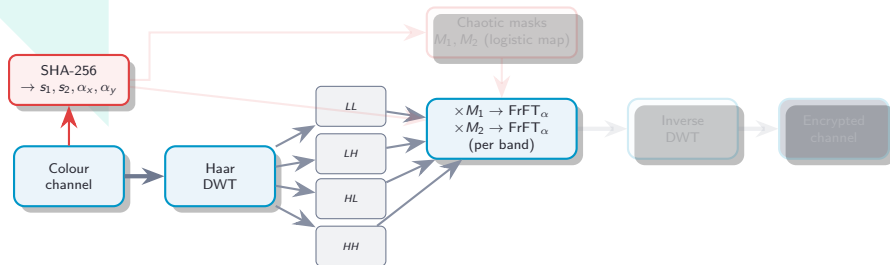
Advanced Chaotic DRPE — Flow



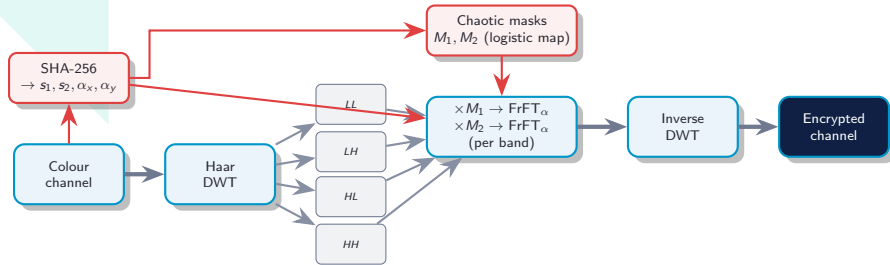
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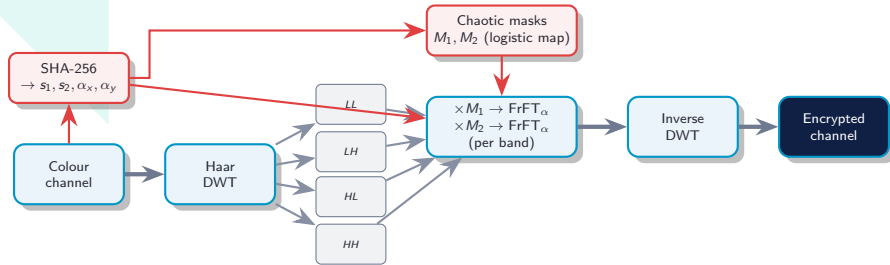


Image-bound keys + chaos + fractional rotation = avalanche effect

Splits image into blurry base +
3 edge layers (Haar filter)

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3 edge layers (Haar filter)

$$x_{n+1} = rx_n(1 - x_n), \quad r = 3.99$$

- tiny seed change $\Rightarrow$ total divergence

Core Signal Concepts



DWT

Splits image into blurry base +
3 edge layers (Haar filter)



FrFT

Rotates signal by *any* decimal
angle, not just 90°



Chaotic Logistic Map

$x_{n+1} = rx_n(1 - x_n)$, $r = 3.99$
— tiny seed change $\Rightarrow$ total
divergence

Attack Simulations

★ Key Error %

Almost-correct password guess

☆ Crop Area %

Data loss / truncated file

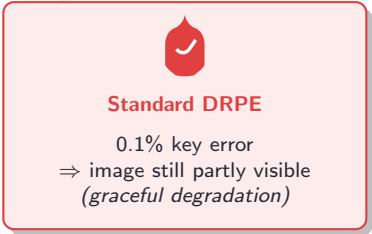
✧ Noise Power %

Noisy channel (AWGN)

◆ Bandwidth %

Narrow-pipe / weak signal

Results: Average



Standard DRPE

0.1% key error
 $\Rightarrow$ image still partly visible
(graceful degradation)



Standard DRPE

0.1% key error
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Results: Avalanche Effect



Standard DRPE

0.1% key error
⇒ image still partly visible
(*graceful degradation*)



Insecure: brute-force feasible



Advanced Chaotic DRPE

0.001% key error
⇒ pure white noise
(*total collapse*)

Results: Avalanche Effect



Standard DRPE

0.1% key error
⇒ image still partly visible
(*graceful degradation*)



Insecure: brute-force feasible



Advanced Chaotic DRPE

0.001% key error
⇒ pure white noise
(*total collapse*)



Secure: infinite decimal key space

Two parallel, wavy lines in a teal color that span the width of the slide, positioned near the top.

03 – IMPACT

Where This Matters

Real-World Applications



Military Drones

Video feeds remain usable through jamming and static



Deep-Space Satellites

Corrupted packets can still preserve the overall image shape



Cancelable Biometrics

Stolen biometric data is useless without the dynamic keys

A series of light blue wavy lines that span horizontally across the upper portion of the slide, positioned behind the central text box.

Linear DRPE: structurally insecure

Linear DRPE: structurally insecure



+ DWT, FrFT, Chaos, SHA-256 keys

```
graph TD; A[Linear DRPE: structurally insecure] --> B["+ DWT, FrFT, Chaos, SHA-256 keys"]; B --> C[Advanced Chaotic DRPE: avalanche-secure];
```

Linear DRPE: structurally insecure

+ DWT, FrFT, Chaos, SHA-256 keys

Advanced Chaotic DRPE: avalanche-secure

Linear DRPE: structurally insecure



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Advanced Chaotic DRPE: avalanche-secure



Thank You